Project Sakura: AI-Powered Clinical Intelligence Platform

Product Requirements Document (PRD)

Document Metadata

- **Version:** 1.1
- **Status:** Draft
- **Date:** July 13, 2026
- **Prepared By:** Person
- **Target Audience:** Engineering, Data Science, Clinical Advisory Board, Product Stakeholders

1. Executive Summary & Strategic Vision

1.1 Product Vision

To serve as the definitive intelligent operating system for clinical decision support, transforming fragmented, multi-modal healthcare data into programmatic, actionable, and verifiable clinical intelligence.

1.2 Mission Statement

To mitigate cognitive overload and diagnostic friction for clinicians by automating the aggregation, serialization, validation, and contextualization of patient information while enforcing complete transparency, data provenance, and evidence traceability.

1.3 Product Positioning

Sakura is an enterprise-grade AI Clinical Intelligence Platform. It functions as an analytical workspace that aggregates fragmented patient data to optimize clinical workflows, surface latent risk vectors, and deliver explainable decision support. It is not a conversational medical chatbot, a direct diagnostic engine, or an Electronic Medical Record (EMR) replacement.

2. Problem Statement & Proposed Solution

2.1 Problem Space

Modern healthcare delivery suffers from severe data fragmentation. Clinician efficiency and patient safety are compromised because critical patient indicators are distributed across isolated document types and legacy systems. Clinicians spend a disproportionate amount of time manually reconciling these sources to establish a comprehensive clinical narrative. Furthermore, standard language modeling solutions fail to validate factual consistency across documents or provide audit trails for generated clinical insights.

2.2 Solution Architecture

Sakura unifies multi-source data streams into a centralized, graph-backed intelligence workspace. The platform applies optical character recognition (OCR), multi-modal entity extraction, and retrieval-augmented generation (RAG) to build longitudinal timelines, perform programmatic consistency checks, and execute natural language queries—all reinforced by a transparent clinical reasoning chain.

3. System Scope & Boundaries

3.1 In-Scope Features (MVP)

- Multi-modal document ingestion and clinical entity extraction.
- Longitudinal, source-linked patient timeline generation.
- Cross-document clinical contradiction and discrepancy detection.
- Clinical data completeness auditing.
- Dynamic trust-scoring for all system-generated insights.
- Graph-backed clinical QA and retrieval-augmented natural language interface.

3.2 Out-of-Scope (Explicit Exclusions)

The platform serves strictly as a passive clinical decision support system and does not possess autonomous execution authority.

- Direct replacement or operational replication of primary EMR systems.

- Patient-facing user interfaces or mobile applications.
- Autonomous automation of medical diagnoses or clinical triage.
- Direct generation or execution of electronic prescriptions.
- Write-back integrations to live production hospital databases during MVP phase.
- Operational administrative features including patient scheduling or billing calculation.

4. User Personas & Access Control

Persona	Role	Primary Objectives	Platform Touchpoints
Attending Physicians	Specialists	Accelerate chart review, identify risks, validate treatment safety.	Patient Workspace, Clinical Alert Engine, AI Assistant.
Resident Doctors	Extenders	Synthesize longitudinal patient histories, conduct pre-rounding reviews.	AI Patient Summary, Interactive Timeline, Document Viewer.
Medical Billing	Insurance Teams	Verify documentation completeness against institutional and payer guidelines.	Insurance Validation Module, Document Intelligence View.
Hospital Administrators	Compliance/Ops	Monitor platform utility metrics and audit trail logs.	Dashboard Analytics, System Administration Console.

5. Functional Requirements

5.1 Identity & Access Management (IAM)

- **REQ-001:** The system must enforce multi-factor authentication (MFA) prior to session initialization.
- **REQ-002:** Role-Based Access Control (RBAC) must dictate document visibility and feature access according to the user persona matrix.

- **REQ-003:** The platform must execute automated session termination after a configurable period of user inactivity.

5.2 Document Intelligence & Ingestion Pipeline

- **REQ-004:** The ingestion pipeline must support file formats including PDF, DICOM text layers, HL7 payloads, and raw text files.
- **REQ-005:** The system must feature a specialized medical OCR engine capable of handling low-resolution or skewed document scans.
- **REQ-006:** The Named Entity Recognition (NER) pipeline must isolate and extract standard clinical vocabulary, including SNOMED-CT codes, ICD-10/11 diagnostic markers, RxNorm ingredients, and LOINC laboratory concepts.

5.3 Core AI & Workspace Features

- **REQ-007: AI Patient Summary:** The system must automatically compile unstructured records into a structured profile containing identified active pathologies, chronic conditions, active prescriptions, verified allergies, and recent critical laboratory values.
- **REQ-008: Interactive Patient Timeline:** The platform must plot all extracted clinical events chronologically.
- **REQ-009: Bidirectional Linking:** Every discrete node on the timeline must maintain a bidirectional deterministic link to its source file coordinates within the Document Viewer.
- **REQ-010: Medical Knowledge Graph:** The system must instantiate semantic relationships within a graph database, mapping dependencies between explicit entities.
- **REQ-011: Medication Safety Engine:** The engine must continuously screen active prescriptions against historical patient records to flag drug-drug interactions, absolute contraindications, documented allergies, and dosage adjustments required by physiological function trends. Additionally, the engine must execute pediatric-specific dosage validation utilizing weight-based calculations and age-adjusted growth curves.
- **REQ-012: Insurance Validation Module:** The system must parse patient profiles against standardized coverage criteria to report document omissions and structural criteria metrics related to claims readiness.

- **REQ-013: AI Clinical Assistant:** The natural language assistant must resolve clinical queries via a Graph-RAG architecture.
- **REQ-014: Source Citation:** Every generated output text block must be appended with explicit, hyperlinked citations referencing the exact source documents utilized to build the answer.

6. The Sakura Clinical Integrity Engine

6.1 Clinical Consistency Engine

- **REQ-015:** The engine must systematically analyze newly ingested records against the existing patient graph to isolate direct contradictions, including opposing diagnostic statements and conflicting immunological reactions.

6.2 Missing Information Detector

- **REQ-016:** The system must evaluate whether the available clinical dataset satisfies baseline diagnostic criteria.
- **REQ-017:** If required clinical parameters are missing, the LLM must suppress speculative assertions and explicitly flag the information gap to the user.

6.3 Trust Score Engine

- **REQ-018:** The platform must calculate a programmatic Trust Score for every generated clinical synthesis or decision support recommendation.
- **REQ-019:** The Trust Score algorithm must be derived dynamically from deterministic metrics: evidence validation level, guideline synchronization, citation density, and underlying data completeness.

6.4 Clinical Alert Engine

- **REQ-020:** The system must trigger immediate asynchronous alert states whenever the integrity engine surfaces high-severity medication interactions or critical documentation gaps.

7. Non-Functional Requirements (NFRs)

7.1 Performance & Latency

- **NFR-001:** Workspace dashboard rendering and interface transitions must load in less than 2.0 seconds.
- **NFR-002:** AI Assistant RAG queries must execute and return complete, cited responses within a maximum limit of 10.0 seconds.

7.2 Security, Privacy & Compliance

- **NFR-003:** All data stores must be encrypted at rest utilizing AES-256 standards.
- **NFR-004:** The system must generate immutable, append-only audit trail logs capturing all user identities and timestamps.
- **NFR-005:** The environment must conform to HIPAA and SOC 2 Type II architectural criteria.
- **NFR-008:** The system must implement Zero Trust Architecture (ZTA) principles, requiring continuous verification of all users and devices regardless of network location or perimeter status.

7.3 Scalability & Architectural Modularity

- **NFR-006:** The AI inference pipeline must operate completely decoupled from the web application backend.
- **NFR-007:** Vector search structures and graph schemas must scale linearly to support multi-million node patient datasets.

8. Success Metrics & KPIs

Metric Category	Target Key Performance Indicator (KPI)	Measurement Methodology
Clinical Efficiency	≥ 40% reduction in chart review velocity.	Time-motion tracking of clinician chart navigation.

Metric Category	Target Key Performance Indicator (KPI)	Measurement Methodology
Safety Capture	100% identification of documented medication contradictions.	Benchmarking against validated synthetic patient corpora.
AI Reliability	0% un-cited assertions or direct hallucinations.	Automated verification of LLM output against source documents.
User Adoption	System Usability Scale (SUS) Score ≥ 80.	Post-trial clinical user survey administration.

9. Technical Architecture & Technology Stack

- **Frontend Interface:** Next.js (Version 14+ App Router), React, TypeScript, Tailwind CSS, shadcn/ui.
- **Application Backend:** FastAPI framework (Python 3.11+), asynchronous event-driven task queues.
- **Relational Storage:** PostgreSQL configured with row-level security policies.
- **Graph Engine:** Amazon Neptune (Serverless) utilizing Gremlin for graph traversal and high-availability clinical relationship mapping.
- **Vector Engine:** Qdrant distributed cluster or pgvector extensions.
- **Orchestration & AI Core:** LangGraph for multi-agent workflows, LlamaIndex for advanced index routing, and enterprise LLM APIs via secure endpoints.
- **Observability:** Langfuse for specialized LLM tracing and evaluation telemetry.